Clinical Natural Language Technology for Health Care: Past, Present, and Future

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Clinical natural language technology refers to the family of computational methods that convert unstructured clinical content—progress notes, discharge summaries, pathology and radiology reports, and scanned or faxed charts—into structured, machine-usable information. It spans classical natural language processing (NLP), optical character recognition (OCR) and computer vision for digitizing documents, and, most recently, large language models (LLMs) and large multimodal models (LMMs) that read text and images jointly. Roughly 80% of healthcare data is unstructured (Nadkarni et al., 2011), and for a payment-integrity and quality-analytics company such as Cotiviti, that text is precisely where coding accuracy, quality gaps, and recoverable dollars are hidden. The ability to read clinical language at scale is therefore a direct lever on Cotiviti's core business of Treatment, Payment, and Operations analytics.

From Rules to Foundation Models

The field has moved through three eras. Early clinical NLP was rule-based: curated dictionaries, regular expressions, ontology lookups against terminologies such as UMLS, SNOMED CT, and ICD-10, and hand-written negation logic (e.g., the NegEx algorithm). Open tools like Apache cTAKES and MetaMap embodied this approach. These systems are transparent and deterministic, but brittle: they only find terms they were told about, generalize poorly across institutions, and require heavy expert maintenance. The present era is defined first by deep learning—domain-tuned transformers such as BioBERT and ClinicalBERT—and now by foundation models. LLMs perform zero- and few-shot extraction, summarization, and coding without task-specific training (Wu et al., 2020; Singhal et al., 2023). In parallel, OCR and computer vision digitize scanned records, and multimodal models fold perception and language into a single step (AlSaad et al., 2024). The emerging future is generalist medical AI: foundation models that ingest notes, images, and structured data together (Moor et al., 2023; Thirunavukarasu et al., 2023), supported by retrieval-augmented generation, ambient documentation, and agentic workflows that chain extraction, validation, and decision support.

Opportunities for Cotiviti

Several opportunities map directly onto Cotiviti's franchise. First, chart-to-code automation for risk adjustment and payment integrity: LLMs can extract diagnoses from notes, map them to ICD-10 and CMS-HCC categories (Centers for Medicare & Medicaid Services, 2024), and flag suspected under- or over-coding for human review, compressing a labor-intensive medical-record review process. Second, content management: summarizing and comparing billing and coding policies and clinical guidelines, and converting written policy into executable rules. Third, automated abstraction of quality measures (e.g., HEDIS) from free text. Fourth, clinical validation for prior authorization and appeals. In each case, the differentiator is not the model alone but evidence-linked, auditable output that a payer can defend to a provider.

Threats and Constraints

The same power introduces risk. LLMs can hallucinate codes or evidence, which is unacceptable when a denial or payment depends on it. Protected health information triggers HIPAA and data-governance obligations, constraining where models may run. Regulators (FDA, the Office of the National Coordinator) are actively scoping clinical AI, and payer-provider disputes demand explainability and auditability that opaque models resist. Model bias can produce inequitable outcomes across populations, and reliance on third-party model vendors raises cost and lock-in concerns. These are governance problems as much as technical ones.

Recommended Strategic Actions

Cotiviti should invest in trustworthy, explainable clinical language AI as a differentiator rather than chase raw model capability. Four concrete actions follow. (1) Build a governed, human-in-the-loop clinical LLM platform for chart review that returns span-level evidence and confidence for every extracted code, making output auditable in payer-provider disputes. (2) Develop a policy-to-rules and extraction pipeline that uses LLMs to draft structured outputs and validates them against ICD-10/HCC terminologies, keeping a deterministic check on a probabilistic model. (3) Adopt multimodal (LMM) models for scanned and faxed records to eliminate brittle, multi-stage OCR pipelines. (4) Stand up an evaluation and guardrail framework —measuring clinical accuracy, bias, and PHI handling—before any production deployment. The accompanying proof-of-concept demonstrates this arc concretely: it turns an unstructured clinical note into codeable ICD-10/HCC findings, contrasting a brittle rule-based pass with LLM and multimodal extraction, and surfacing evidence and confidence for each finding. Positioned this way, clinical language technology becomes a defensible, compounding advantage for Cotiviti rather than a commoditized feature.

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